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Bayesian Statistics and Data Analysis

Lecture 3

Måns Magnusson

Department of Statistics, Uppsala University
Thanks to Aki Vehtari, Aalto University

- Introduction
- Multiple parameter models
 - Marginalization
- Gaussian Model
 - Example
 - Conjugate prior
- Bioassay example
- Additional examples
 - Multinomial model
 - Multivariate Gaussian



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Assignment 2

- **Introduction**
- **Multiple parameter models**
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 - Multinomial model
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- “Tasks 2b and 2d were unclear” (on our list now)



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- “Tasks 2b and 2d were unclear” (on our list now)
- “Simulate from the predictive distribution in two steps?”



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- “Tasks 2b and 2d were unclear” (on our list now)
- “Simulate from the predictive distribution in two steps?”



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Practicalities

- **Introduction**
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- **Bioassay example**
- **Additional examples**
 - Multinomial model
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- **Master thesis proposals**
 - Deadline for interest today at 23.59.
 - I'll stay if you have questions.



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Practicalities

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- Master thesis proposals

- Deadline for interest today at 23.59.
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- Next week (Wednesday): Deadline mini-project proposal



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Practicalities

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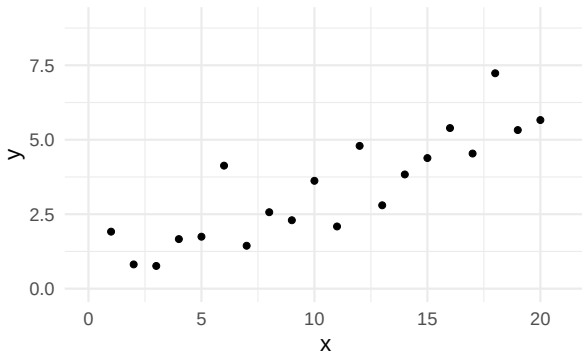
- Multivariate Gaussian

- Master thesis proposals
 - Deadline for interest today at 23.59.
 - I'll stay if you have questions.
- Next week (Wednesday): Deadline mini-project proposal
- Submit to the correct assignment - the submission is anonymous



Example of uncertainty in modeling

Data

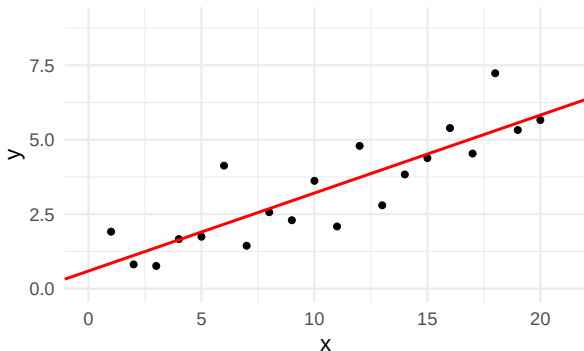


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Example of uncertainty in modeling

Posterior mean

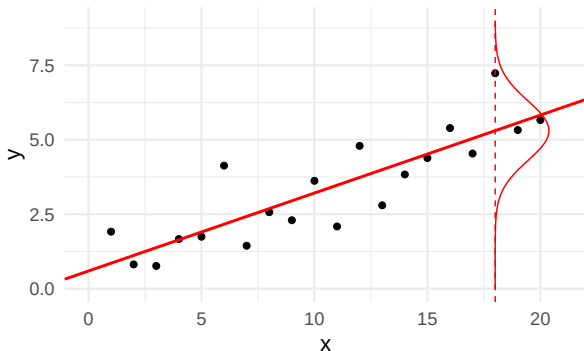


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Example of uncertainty in modeling

Predictive distribution given posterior mean

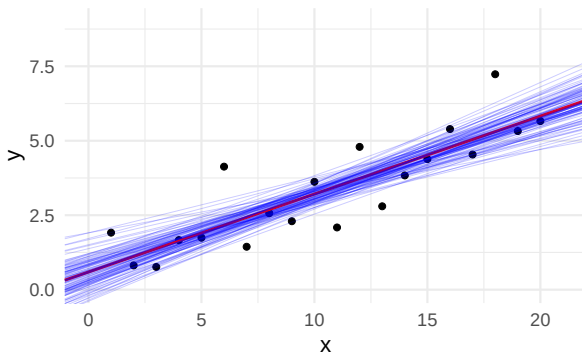


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Example of uncertainty in modeling

Posterior draws

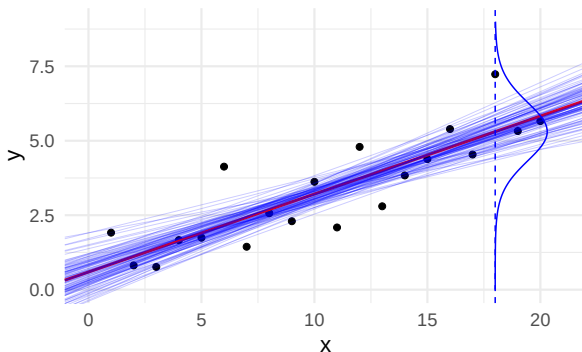


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Example of uncertainty in modeling

Posterior draws and predictive distribution

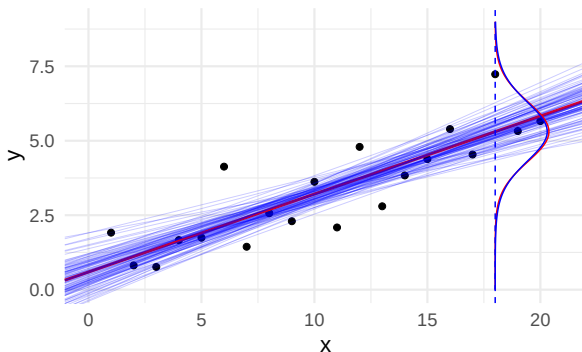


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Example of uncertainty in modeling

Posterior draws and predictive distribution



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Monte Carlo and Posterior Draws

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- Assume we can get draws from $p(\theta | y)$



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- Assume we can get draws from $p(\theta | y)$
- $\theta^{(s)}$ draws from $p(\theta | y)$ can be used
 - for visualization
 - to approximate expectations (integrals)

$$E_{p(\theta|y)}[\theta] = \int \theta p(\theta | y) \approx \frac{1}{S} \sum_{s=1}^S \theta^{(s)}$$



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- Assume we can get draws from $p(\theta | y)$
- $\theta^{(s)}$ draws from $p(\theta | y)$ can be used
 - for visualization
 - to approximate expectations (integrals)

$$E_{p(\theta|y)}[\theta] = \int \theta p(\theta | y) \approx \frac{1}{S} \sum_{s=1}^S \theta^{(s)}$$

- to approximate uncertainty intervals for θ



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Section 2

Multiple parameter models



- Joint posterior distribution of multiple parameters

$$p(\theta_1, \theta_2 \mid y) \propto p(y \mid \theta_1, \theta_2)p(\theta_1, \theta_2)$$

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- Joint posterior distribution of multiple parameters

$$p(\theta_1, \theta_2 | y) \propto p(y | \theta_1, \theta_2)p(\theta_1, \theta_2)$$

- Marginalization

$$p(\theta_1 | y) = \int p(\theta_1, \theta_2 | y) d\theta_2$$

$p(\theta_1 | y)$ is a marginal distribution

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- Joint posterior distribution of multiple parameters

$$p(\theta_1, \theta_2 \mid y) \propto p(y \mid \theta_1, \theta_2)p(\theta_1, \theta_2)$$

- Marginalization

$$p(\theta_1 \mid y) = \int p(\theta_1, \theta_2 \mid y) d\theta_2$$

$p(\theta_1 \mid y)$ is a marginal distribution

- Goal is often to find marginal posterior of an interesting quantity
 - a **parameter** $p(\theta|y)$
 - a **potential observation** $p(\tilde{y}|y)$

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- Joint distribution of unknown future observation and parameters

$$\begin{aligned} p(\tilde{y}, \theta | y) &= p(\tilde{y} | \theta, y) p(\theta | y) \\ &= p(\tilde{y} | \theta) p(\theta | y) \quad (\text{often}) \end{aligned}$$

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- Joint distribution of unknown future observation and parameters

$$\begin{aligned} p(\tilde{y}, \theta | y) &= p(\tilde{y} | \theta, y) p(\theta | y) \\ &= p(\tilde{y} | \theta) p(\theta | y) \quad (\text{often}) \end{aligned}$$

- Marginalization over posterior distribution

$$\begin{aligned} p(\tilde{y} | y) &= \int p(\tilde{y} | \theta) p(\theta | y) d\theta \\ &= \int p(\tilde{y}, \theta | y) d\theta \end{aligned}$$

$p(\tilde{y} | y)$ is a predictive distribution

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- Observation model

$$\frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(y - \mu)^2\right)$$

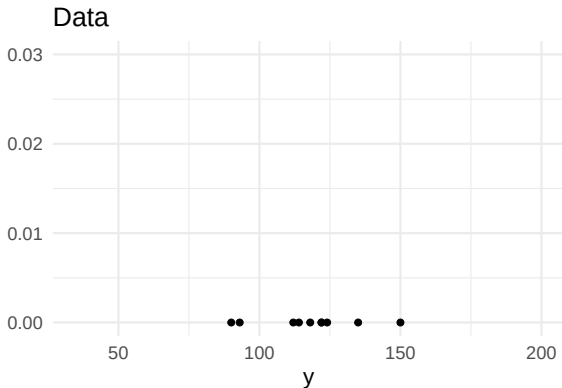
- Uninformative prior

$$p(\mu, \sigma^2) \propto \sigma^{-2}$$



Gaussian example

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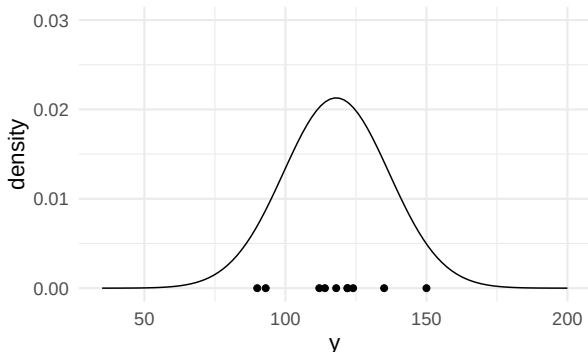




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Gaussian example

Gaussian fit with posterior mean



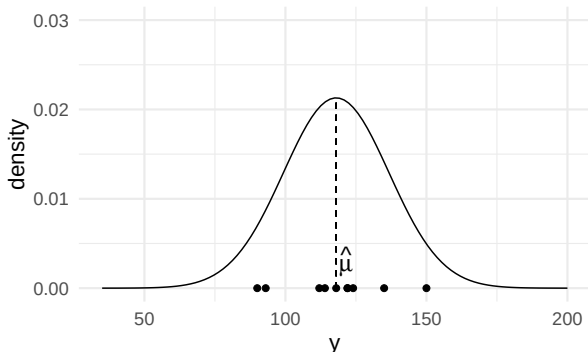
$$p(\textcolor{red}{y} \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(\textcolor{red}{y} - \mu)^2\right)$$



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Gaussian example

Gaussian fit with posterior mean



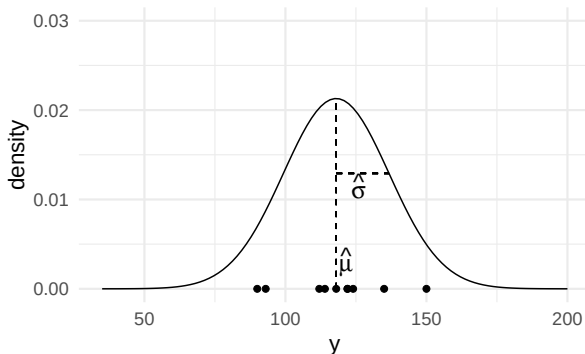
$$p(y \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(y - \mu)^2\right)$$



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Gaussian example

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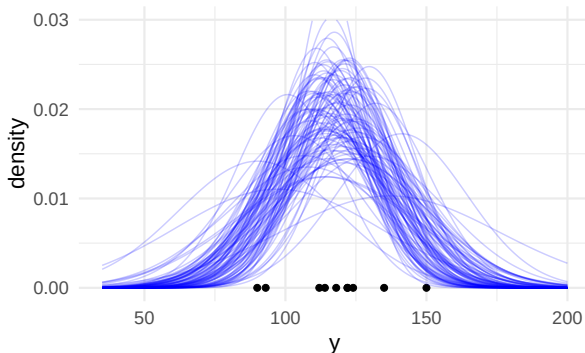
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Gaussian example

Gaussians with posterior draw parameters



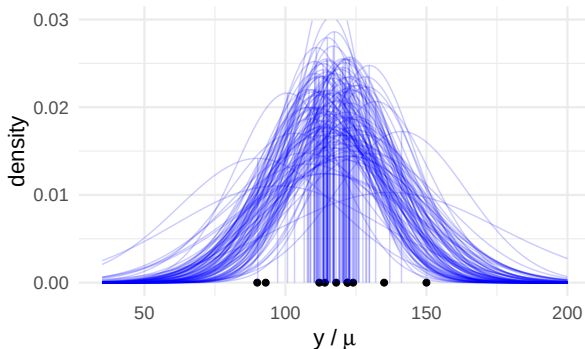
$$\mu^{(s)}, \sigma^{(s)} \sim p(\mu, \sigma \mid y)$$



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Gaussian example

Gaussians with posterior draw parameters



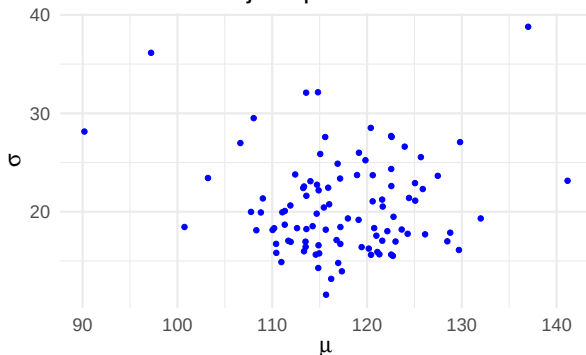
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Gaussian example

Draws from the joint posterior distribution



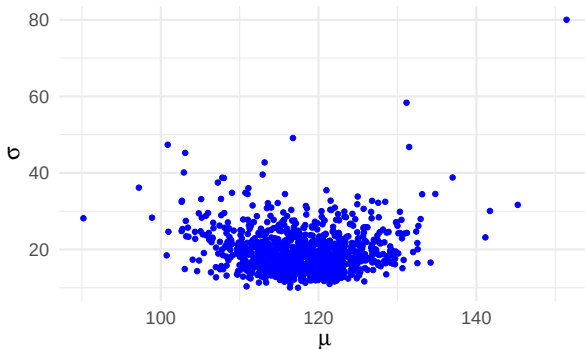
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Gaussian example

Draws from the joint posterior distribution

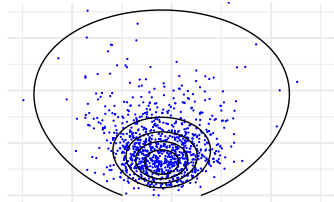


$$\mu^{(s)}, \sigma^{(s)} \sim p(\mu, \sigma \mid y)$$



Joint posterior

$$\mu^{(s)}, \sigma^{(s)} \sim p(\mu, \sigma \mid y)$$



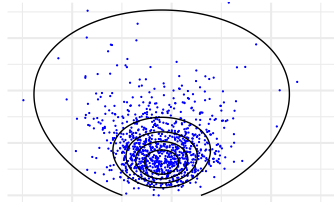
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Joint posterior

$$\mu^{(s)}, \sigma^{(s)} \sim p(\mu, \sigma \mid y)$$

with $p(\mu, \sigma^2) \propto \sigma^{-2}$



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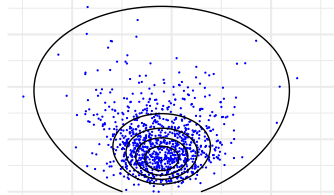
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Joint posterior

$$\mu^{(s)}, \sigma^{(s)} \sim p(\mu, \sigma | y)$$

with $p(\mu, \sigma^2) \propto \sigma^{-2}$

$$p(\mu, \sigma^2 | y) \propto \sigma^{-2} \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{1}{2\sigma^2}(y_i - \mu)^2\right)$$





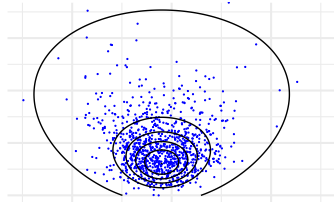
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Joint posterior

$$\mu^{(s)}, \sigma^{(s)} \sim p(\mu, \sigma | y)$$

with $p(\mu, \sigma^2) \propto \sigma^{-2}$

$$p(\mu, \sigma^2 | y) \propto \sigma^{-n-2} \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2 \right)$$





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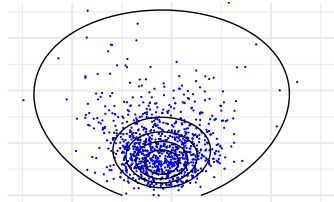
Joint posterior

$$\mu^{(s)}, \sigma^{(s)} \sim p(\mu, \sigma | y)$$

with $p(\mu, \sigma^2) \propto \sigma^{-2}$

$$\begin{aligned} p(\mu, \sigma^2 | y) &\propto \sigma^{-n-2} \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2 \right) \\ &= \sigma^{-n-2} \exp \left(-\frac{1}{2\sigma^2} \left[\sum_{i=1}^n (y_i - \bar{y})^2 + n(\bar{y} - \mu)^2 \right] \right) \end{aligned}$$

$$\text{where } \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$



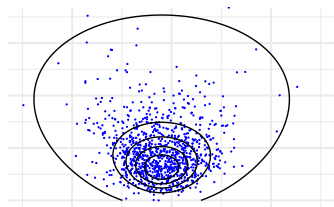


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Joint posterior

$$\mu^{(s)}, \sigma^{(s)} \sim p(\mu, \sigma | y)$$

with $p(\mu, \sigma^2) \propto \sigma^{-2}$



$$\begin{aligned} p(\mu, \sigma^2 | y) &\propto \sigma^{-n-2} \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2 \right) \\ &= \sigma^{-n-2} \exp \left(-\frac{1}{2\sigma^2} \left[\sum_{i=1}^n (y_i - \bar{y})^2 + n(\bar{y} - \mu)^2 \right] \right) \end{aligned}$$

$$\text{where } \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

$$= \sigma^{-n-2} \exp \left(-\frac{1}{2\sigma^2} [(n-1)s^2 + n(\bar{y} - \mu)^2] \right)$$

$$\text{where } s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$$



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Gaussian: Completing the square

$$\sum_{i=1}^n (y_i - \mu)^2$$

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$$\sum_{i=1}^n (y_i - \mu)^2$$
$$\sum_{i=1}^n (y_i^2 - 2y_i\mu + \mu^2)$$

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Gaussian: Completing the square

$$\sum_{i=1}^n (y_i - \mu)^2$$

$$\sum_{i=1}^n (y_i^2 - 2y_i\mu + \mu^2)$$

$$\sum_{i=1}^n (y_i^2 - 2y_i\mu + \mu^2 - \bar{y}^2 + \bar{y}^2 - 2y_i\bar{y} + 2y_i\bar{y})$$



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Gaussian: Completing the square

$$\sum_{i=1}^n (y_i - \mu)^2$$

$$\sum_{i=1}^n (y_i^2 - 2y_i\mu + \mu^2)$$

$$\sum_{i=1}^n (y_i^2 - 2y_i\mu + \mu^2 - \bar{y}^2 + \bar{y}^2 - 2y_i\bar{y} + 2y_i\bar{y})$$

$$\sum_{i=1}^n (y_i^2 - 2y_i\bar{y} + \bar{y}^2) + \sum_{i=1}^n (\mu^2 - 2y_i\mu - \bar{y}^2 + 2y_i\bar{y})$$



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Gaussian: Completing the square

$$\sum_{i=1}^n (y_i - \mu)^2$$

$$\sum_{i=1}^n (y_i^2 - 2y_i\mu + \mu^2)$$

$$\sum_{i=1}^n (y_i^2 - 2y_i\mu + \mu^2 - \bar{y}^2 + \bar{y}^2 - 2y_i\bar{y} + 2y_i\bar{y})$$

$$\sum_{i=1}^n (y_i^2 - 2y_i\bar{y} + \bar{y}^2) + \sum_{i=1}^n (\mu^2 - 2y_i\mu - \bar{y}^2 + 2y_i\bar{y})$$

$$\sum_{i=1}^n (y_i - \bar{y})^2 + n(\mu^2 - 2\bar{y}\mu - \bar{y}^2 + 2\bar{y}\bar{y})$$



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Gaussian: Completing the square

$$\sum_{i=1}^n (y_i - \mu)^2$$

$$\sum_{i=1}^n (y_i^2 - 2y_i\mu + \mu^2)$$

$$\sum_{i=1}^n (y_i^2 - 2y_i\mu + \mu^2 - \bar{y}^2 + \bar{y}^2 - 2y_i\bar{y} + 2y_i\bar{y})$$

$$\sum_{i=1}^n (y_i^2 - 2y_i\bar{y} + \bar{y}^2) + \sum_{i=1}^n (\mu^2 - 2y_i\mu - \bar{y}^2 + 2y_i\bar{y})$$

$$\sum_{i=1}^n (y_i - \bar{y})^2 + n(\mu^2 - 2\bar{y}\mu - \bar{y}^2 + 2\bar{y}\bar{y})$$

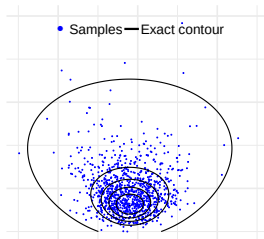
$$\sum_{i=1}^n (y_i - \bar{y})^2 + n(\bar{y} - \mu)^2$$



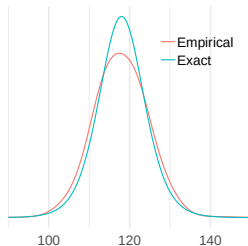
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Marginal $p(\mu)$ and $p(\sigma^2)$

Joint posterior



Marginal of mu



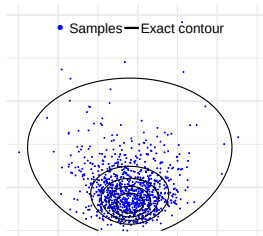
$$\mu^{(s)}, \sigma^{(s)} \sim p(\mu, \sigma \mid y)$$



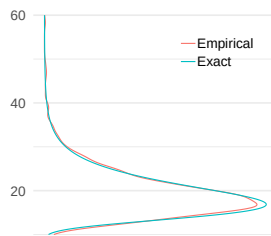
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Marginal $p(\mu)$ and $p(\sigma^2)$

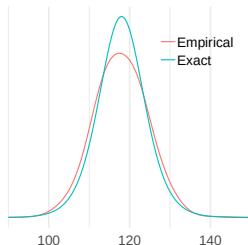
Joint posterior



Marginal of sigma



Marginal of mu



$$\mu^{(s)}, \sigma^{(s)} \sim p(\mu, \sigma | y)$$

marginals

$$p(\mu | y) = \int p(\mu, \sigma | y) d\sigma$$

$$p(\sigma | y) = \int p(\mu, \sigma | y) d\mu$$



Pair discussion: nuisance parameters

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- Suppose we care about μ in the Gaussian model.
 - Which parameter is a nuisance parameter?
 - Why integrate it out instead of plug in one estimate?



Pair discussion: nuisance parameters

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- Suppose we care about μ in the Gaussian model.
- Which parameter is a nuisance parameter?
- Why integrate it out instead of plug in one estimate?

After discussion:

- σ (or σ^2) is nuisance if the target is μ .
- Marginalization keeps uncertainty about σ in $p(\mu | y)$.
- Plug-in conditions on one value and usually understates uncertainty.



Marginal $p(\mu|y)$ and $p(\sigma^2|y)$

Marginal posterior $p(\sigma^2 | y)$ (easier for σ^2 than σ)

$$p(\sigma^2 | y) \propto \int p(\mu, \sigma^2 | y) d\mu$$

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Marginal posterior $p(\sigma^2 | y)$ (easier for σ^2 than σ)

$$\begin{aligned} p(\sigma^2 | y) &\propto \int p(\mu, \sigma^2 | y) d\mu \\ &\propto \int \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2} [(n-1)s^2 + n(\bar{y} - \mu)^2]\right) d\mu \end{aligned}$$



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Note! $\int \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2} (y - \mu)^2\right) d\mu = 1$



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Note!

$$\begin{aligned}&\int \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2} (y - \mu)^2\right) d\mu = 1 \\&\propto \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2} (n-1)s^2\right) \sqrt{2\pi\sigma^2/n}\end{aligned}$$



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$$p(\sigma^2 | y) = \text{Inv-}\chi^2(\sigma^2 | n-1, s^2)$$



Known mean

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$$\sigma^2 \mid y \sim \text{Inv-}\chi^2(n, \nu)$$

$$\text{where } \nu = \frac{1}{n} \sum_{i=1}^n (y_i - \mu)^2$$



Known mean

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$$\sigma^2 \mid y \sim \text{Inv-}\chi^2(n, \nu)$$

$$\text{where } \nu = \frac{1}{n} \sum_{i=1}^n (y_i - \mu)^2$$

Unknown mean

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$$\sigma^2 \mid y \sim \text{Inv-}\chi^2(n - 1, s^2)$$

$$\text{where } s^2 = \frac{1}{n - 1} \sum_{i=1}^n (y_i - \bar{y})^2$$



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- Marginal posterior $p(\mu | y)$

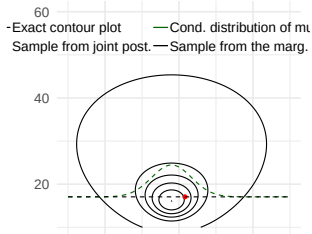
$$p(\mu | y) = \int_0^\infty p(\mu | \sigma^2, y) p(\sigma^2 | y) d\sigma^2$$

- Marginal posterior of μ , a mixture of normal distributions where mixing density is the marginal posterior of σ^2

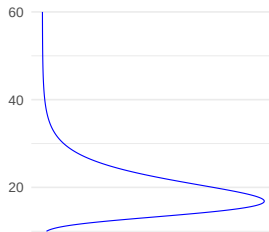


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Joint posterior



Marginal of sigma



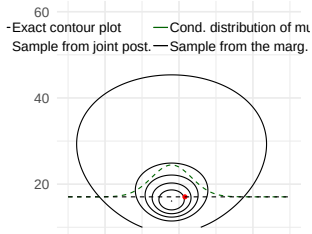
Factorization

$$p(\mu, \sigma^2 | y) = p(\mu | \sigma^2, y) p(\sigma^2 | y)$$

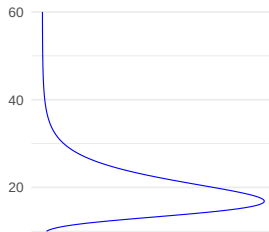


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Joint posterior



Marginal of sigma



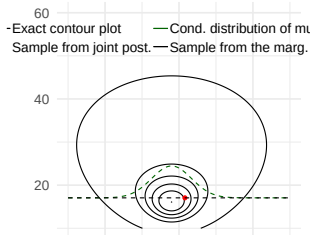
Factorization

$$p(\mu, \sigma^2 | y) = p(\mu | \sigma^2, y) p(\sigma^2 | y)$$
$$(\sigma^2)^{(s)} \sim p(\sigma^2 | y)$$

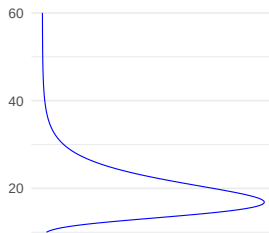


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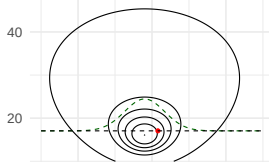
$$p(\mu | (\sigma^2)^{(s)}, y) = \mathcal{N}(\mu | \bar{y}, (\sigma^2)^{(s)} / n)$$



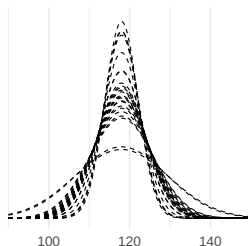
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Joint posterior

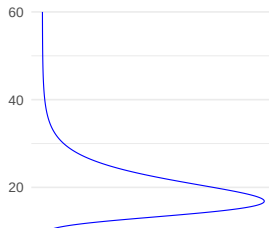
60
-Exact contour plot — Cond. distribution of μ
Sample from joint post. — Sample from the marg.



Cond distr of μ for 25 draws



Marginal of sigma



Factorization

$$p(\mu, \sigma^2 | y) = p(\mu | \sigma^2, y) p(\sigma^2 | y)$$

$$(\sigma^2)^{(s)} \sim p(\sigma^2 | y)$$

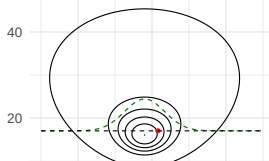
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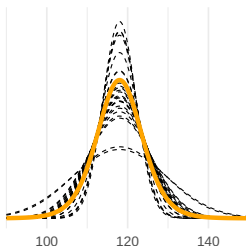
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Joint posterior

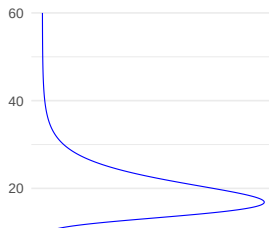
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Cond distr of μ for 25 draws



Marginal of sigma



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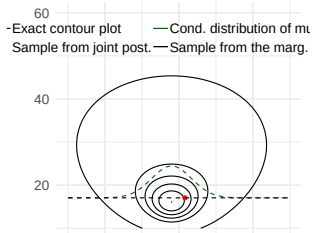
$$p(\mu | (\sigma^2)^{(s)}, y) = \mathcal{N}(\mu | \bar{y}, (\sigma^2)^{(s)} / n)$$

$$p(\mu | y) \approx \frac{1}{S} \sum_{s=1}^S \mathcal{N}(\mu | \bar{y}, (\sigma^2)^{(s)} / n)$$

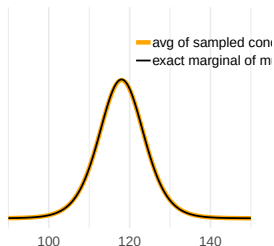


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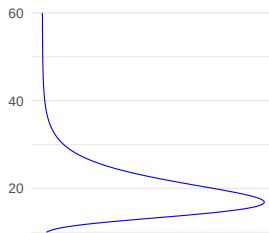
Joint posterior



Cond. distr of μ



Marginal of sigma



Factorization

$$p(\mu, \sigma^2 | y) = p(\mu | \sigma^2, y) p(\sigma^2 | y)$$

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$$p(\mu | y) \approx \frac{1}{S} \sum_{s=1}^S \mathcal{N}(\mu | \bar{y}, (\sigma^2)^{(s)} / n)$$



Marginal posterior $p(\mu \mid y)$

$$p(\mu \mid y) = \int_0^\infty p(\mu, \sigma^2 \mid y) d\sigma^2$$

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$$\begin{aligned} p(\mu \mid y) &= \int_0^\infty p(\mu, \sigma^2 \mid y) d\sigma^2 \\ &\propto \int_0^\infty \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2} [(n-1)s^2 + n(\bar{y} - \mu)^2]\right) d\sigma^2 \end{aligned}$$



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Transformation (integration by substitution)

$$A = (n-1)s^2 + n(\mu - \bar{y})^2$$



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Transformation (integration by substitution)

$$A = (n-1)s^2 + n(\mu - \bar{y})^2 \quad \text{and} \quad z = \frac{A}{2\sigma^2}$$

$$dz = \left(-\frac{A}{2(\sigma^2)^2}\right) d\sigma^2$$



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$$p(\mu \mid y) \propto A^{-n/2} \int_0^\infty z^{(n-2)/2} \exp(-z) dz$$



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Recognize gamma integral $\Gamma(u) = \int_0^\infty x^{u-1} \exp(-x) dx$



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Recognize gamma integral $\Gamma(u) = \int_0^\infty x^{u-1} \exp(-x) dx$

$$\propto [(n-1)s^2 + n(\mu - \bar{y})^2]^{-n/2}$$



$$p(\mu|y) \propto [(n-1)s^2 + n(\mu - \bar{y})^2]^{-n/2}$$

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$$\begin{aligned} p(\mu|y) &\propto [(n-1)s^2 + n(\mu - \bar{y})^2]^{-n/2} \\ &\propto \left[1 + \frac{n(\mu - \bar{y})^2}{(n-1)s^2} \right]^{-n/2} \end{aligned}$$

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$$p(x|\mu_*, \sigma_*, \nu_*) = \frac{\Gamma(\frac{\nu_*+1}{2})}{\sqrt{\pi\nu_*\sigma_*^2} \Gamma(\frac{\nu_*}{2})} \left(1 + \frac{1}{\nu_*} \frac{(x - \mu_*)^2}{\sigma_*^2} \right)^{-\frac{\nu_*+1}{2}}$$

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$$p(\mu|y) \propto [(n-1)s^2 + n(\mu - \bar{y})^2]^{-n/2}$$

$$\propto \left[1 + \frac{n(\mu - \bar{y})^2}{(n-1)s^2} \right]^{-n/2}$$

$$p(x|\mu_*, \sigma_*, \nu_*) = \frac{\Gamma(\frac{\nu_*+1}{2})}{\sqrt{\pi\nu_*\sigma_*^2} \Gamma(\frac{\nu_*}{2})} \left(1 + \frac{1}{\nu_*} \frac{(x - \mu_*)^2}{\sigma_*^2} \right)^{-\frac{\nu_*+1}{2}}$$

$$p(\mu \mid y) = \underbrace{t_{n-1}(\mu \mid \bar{y}, s^2/n)}_{\text{Student's } t}$$



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- For a new observation $\tilde{y}$, which uncertainties should be included?
- uncertainty about μ ?
- noise in a new observation?
- both?
- What are the two types of uncertainties called?



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Pair discussion: future observations

- For a new observation $\tilde{y}$, which uncertainties should be included?
- uncertainty about μ ?
- noise in a new observation?
- both?
- What are the two types of uncertainties called?

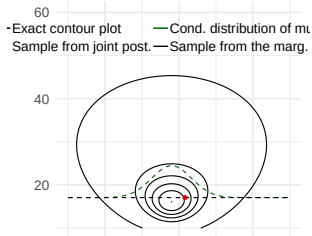
After discussion:

- Parameter uncertainty is epistemic uncertainty.
- New observation noise is aleatoric uncertainty.
- The posterior predictive distribution includes both.

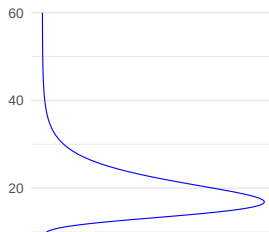


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Joint posterior



Marginal of sigma



Predictive distribution for new $\tilde{y}$

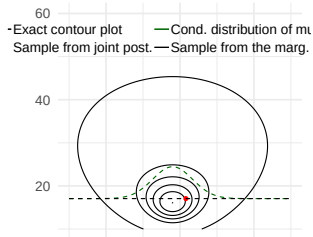
$$p(\tilde{y}|y) = \int p(\tilde{y}|\mu, \sigma) p(\mu, \sigma|y) d\mu d\sigma$$



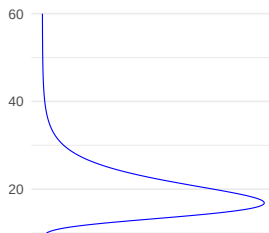
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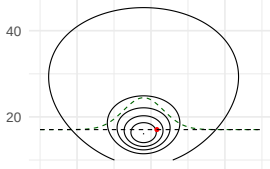
$$\mu^{(s)}, \sigma^{(s)} \sim p(\mu, \sigma|y)$$



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Joint posterior

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-Exact contour plot — Cond. distribution of μ
Sample from joint post. — Sample from the marg.



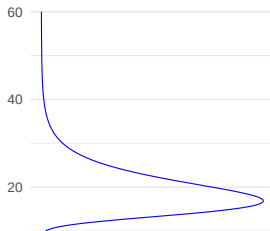
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$$\mu^{(s)}, \sigma^{(s)} \sim p(\mu, \sigma|y)$$

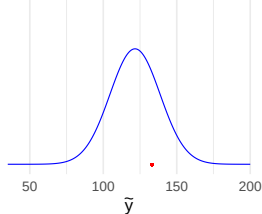
$$\tilde{y}^{(s)} \sim p(\tilde{y}|\mu^{(s)}, \sigma^{(s)})$$

Marginal of sigma



Posterior predictive distribution

- Sample from the predictive distribution
- Predictive distribution given the posterior sample

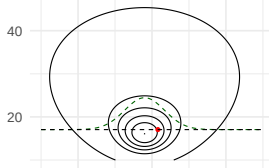




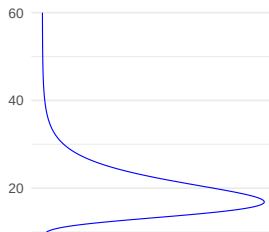
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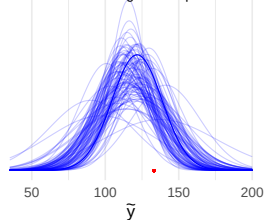
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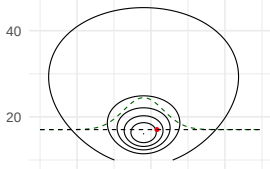




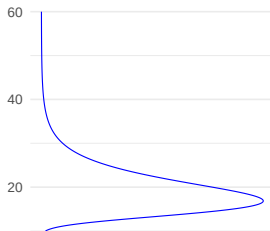
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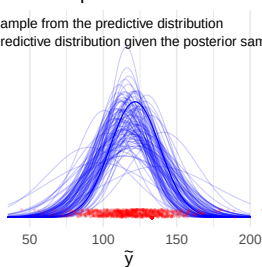
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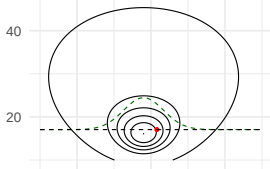




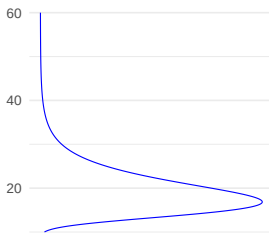
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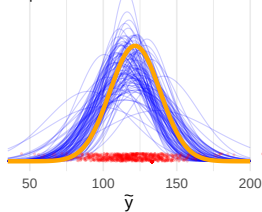
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$$\mu^{(s)}, \sigma^{(s)} \sim p(\mu, \sigma|y)$$

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Posterior predictive distribution

- Sample from the predictive distribution
- Predictive distribution given the posterior sample
- Exact predictive distribution





First condition on σ^2 , then integrate it out

$$p(\tilde{y} \mid \sigma^2, y) = \int p(\tilde{y} \mid \mu, \sigma^2) p(\mu \mid \sigma^2, y) d\mu$$

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First condition on σ^2 , then integrate it out

$$\begin{aligned} p(\tilde{y} \mid \sigma^2, y) &= \int p(\tilde{y} \mid \mu, \sigma^2) p(\mu \mid \sigma^2, y) d\mu \\ &= \int \mathcal{N}(\tilde{y} \mid \mu, \sigma^2) \mathcal{N}(\mu \mid \bar{y}, \sigma^2/n) d\mu \end{aligned}$$

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$$\begin{aligned} p(\tilde{y} \mid \sigma^2, y) &= \int p(\tilde{y} \mid \mu, \sigma^2) p(\mu \mid \sigma^2, y) d\mu \\ &= \int \mathcal{N}(\tilde{y} \mid \mu, \sigma^2) \mathcal{N}(\mu \mid \bar{y}, \sigma^2/n) d\mu \\ &= \mathcal{N}(\tilde{y} \mid \bar{y}, (1 + \frac{1}{n})\sigma^2) \end{aligned}$$

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this is up to scaling factor same as $p(\mu \mid \sigma^2, y)$

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this is up to scaling factor same as $p(\mu \mid \sigma^2, y)$

$$p(\tilde{y} \mid y) = t_{n-1}(\tilde{y} \mid \bar{y}, (1 + \frac{1}{n})s^2)$$

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With $p(\mu, \sigma^2) \propto \sigma^{-2}$ and $s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$:

$$\sigma^2 \mid y \sim \text{Inv-}\chi^2(n-1, s^2)$$

$$\mu \mid y \sim t_{n-1}(\cdot \mid \bar{y}, s^2/n)$$

$$\tilde{y} \mid y \sim t_{n-1}(\cdot \mid \bar{y}, (1 + \frac{1}{n})s^2)$$

- Posterior uncertainty for μ is smaller than predictive uncertainty for a new $\tilde{y}$.



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- In these slides, $t_\nu(\cdot \mid m, \nu)$ uses ν as scale squared.
- If software expects a scale parameter, use $\sqrt{\nu}$.
- Examples:

$$\mu \mid y \sim t_{n-1}(\cdot \mid \bar{y}, s^2/n)$$

$$\tilde{y} \mid y \sim t_{n-1}(\cdot \mid \bar{y}, (1 + \frac{1}{n})s^2)$$



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- Often the quantity of interest is $q = g(\theta)$.
- If $\theta^{(s)} \sim p(\theta | y)$, then compute

$$q^{(s)} = g(\theta^{(s)}).$$

- The draws $q^{(s)}$ summarize $p(q | y)$.
- Examples: $\theta_1 - \theta_2$, $\mu_1 - \mu_2$, $x_{LD50} = -\alpha/\beta$.



Two independent proportions: odds ratio

- For groups $j = 0, 1$:

$$y_j \mid p_j \sim \text{Bin}(n_j, p_j), \quad p_j \sim \text{Beta}(a_j, b_j)$$

- Conjugacy gives

$$p_j \mid y_j \sim \text{Beta}(a_j + y_j, b_j + n_j - y_j)$$

- Draw $p_0^{(s)}$ and $p_1^{(s)}$ independently, then compute

$$\text{OR}^{(s)} = \frac{p_1^{(s)} / (1 - p_1^{(s)})}{p_0^{(s)} / (1 - p_0^{(s)})}.$$

- Summarize the draws with a point estimate, posterior interval, histogram, and a prior sensitivity check.

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Two independent normal means

- For groups $j = 1, 2$:

$$y_{ij} \mid \mu_j, \sigma_j^2 \sim \mathcal{N}(\mu_j, \sigma_j^2), \quad p(\mu_j, \sigma_j^2) \propto \sigma_j^{-2}$$

- From the one-sample result:

$$\mu_j \mid y_j \sim t_{n_j-1}(\cdot \mid \bar{y}_j, s_j^2/n_j)$$

- Draw $\mu_1^{(s)}$ and $\mu_2^{(s)}$ independently, then compute

$$\mu_d^{(s)} = \mu_1^{(s)} - \mu_2^{(s)}.$$

- Summarize $p(\mu_d \mid y)$ with a point estimate, posterior interval, histogram, and probabilities of scientifically meaningful differences.



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- Suppose $\mu_d = \mu_1 - \mu_2$ has a continuous posterior.
 - What is $\Pr(\mu_d = 0 \mid y)$?
 - What would be a more useful posterior summary?



Pair discussion: continuous posteriors

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- Suppose $\mu_d = \mu_1 - \mu_2$ has a continuous posterior.
- What is $\Pr(\mu_d = 0 \mid y)$?
- What would be a more useful posterior summary?

After discussion:

- $\Pr(\mu_d = 0 \mid y) = 0$ for a continuous posterior.
- More useful: $\Pr(\mu_d > 0 \mid y)$ and a credible interval.



Simon Newcomb's speed of light experiment in 1882

Newcomb measured ($n = 66$) the time required for light to travel from his laboratory on the Potomac River to a mirror at the base of the Washington Monument and back, a total distance of 7442 meters.

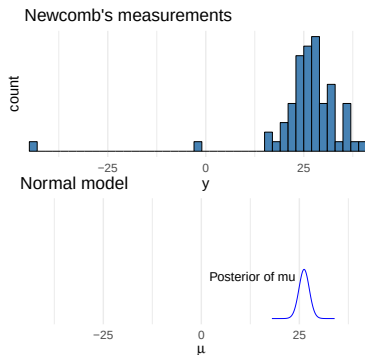
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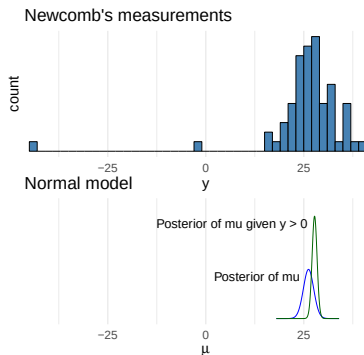




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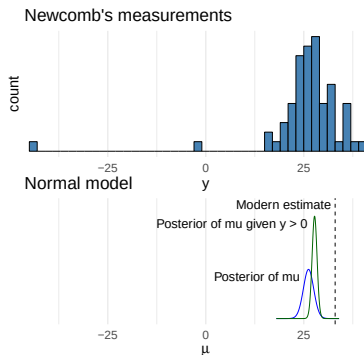




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Newcomb: model checking

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- The Gaussian model is useful as a first analysis.
 - The negative coded values hint at outliers or model misfit.
 - Later: check whether the model can generate data that look like what we observed.



Gaussian - conjugate prior

- Conjugate prior on the form $p(\sigma^2)p(\mu \mid \sigma^2)$

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- Conjugate prior on the form $p(\sigma^2)p(\mu \mid \sigma^2)$
- Handy parameterization

$$\mu \mid \sigma^2 \sim N(\mu_0, \sigma^2 / \kappa_0)$$

$$\sigma^2 \sim \text{Inv-}\chi^2(\nu_0, \sigma_0^2)$$

which can be written as

$$p(\mu, \sigma^2) = N\text{Inv-}\chi^2(\mu_0, \sigma_0^2 / \kappa_0; \nu_0, \sigma_0^2)$$

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$$p(\mu, \sigma^2) = N\text{Inv-}\chi^2(\mu_0, \sigma_0^2 / \kappa_0; \nu_0, \sigma_0^2)$$

- μ and σ^2 are a priori dependent
 - if σ^2 is large, then μ has wide prior



Joint posterior (exercise 3.9)

$$p(\mu, \sigma^2 | y) = \text{N} \text{H} \nu - \chi^2(\mu_n, \sigma_n^2 / \kappa_n; \nu_n, \sigma_n^2)$$

where

$$\mu_n = \frac{\kappa_0}{\kappa_0 + n} \mu_0 + \frac{n}{\kappa_0 + n} \bar{y}$$

$$\kappa_n = \kappa_0 + n$$

$$\nu_n = \nu_0 + n$$

$$\nu_n \sigma_n^2 = \nu_0 \sigma_0^2 + (n - 1) s^2 + \frac{\kappa_0 n}{\kappa_0 + n} (\bar{y} - \mu_0)^2$$

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- Conditional $p(\mu \mid \sigma^2, y)$

$$\begin{aligned}\mu \mid \sigma^2, y &\sim N(\mu_n, \sigma^2 / \kappa_n) \\ &= N\left(\frac{\frac{\kappa_0}{\sigma^2} \mu_0 + \frac{n}{\sigma^2} \bar{y}}{\frac{\kappa_0}{\sigma^2} + \frac{n}{\sigma^2}}, \frac{1}{\frac{\kappa_0}{\sigma^2} + \frac{n}{\sigma^2}}\right)\end{aligned}$$

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- Marginal $p(\sigma^2 \mid y)$

$$\sigma^2 \mid y \sim \text{Inv-}\chi^2(\nu_n, \sigma_n^2)$$



- Conditional $p(\mu \mid \sigma^2, y)$

$$\begin{aligned}\mu \mid \sigma^2, y &\sim N(\mu_n, \sigma^2 / \kappa_n) \\ &= N\left(\frac{\frac{\kappa_0}{\sigma^2} \mu_0 + \frac{n}{\sigma^2} \bar{y}}{\frac{\kappa_0}{\sigma^2} + \frac{n}{\sigma^2}}, \frac{1}{\frac{\kappa_0}{\sigma^2} + \frac{n}{\sigma^2}}\right)\end{aligned}$$

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$$\sigma^2 \mid y \sim \text{Inv-}\chi^2(\nu_n, \sigma_n^2)$$

- Marginal $p(\mu \mid y)$

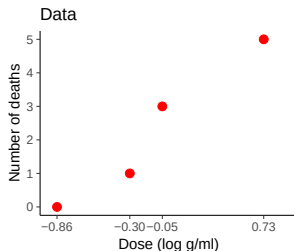
$$\mu \mid y \sim t_{\nu_n}(\mu \mid \mu_n, \sigma_n^2 / \kappa_n)$$

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Bioassay

Dose, x_i (log g/ml)	Number of animals, n_i	Number of deaths, y_i
-0.86	5	0
-0.30	5	1
-0.05	5	3
0.73	5	5



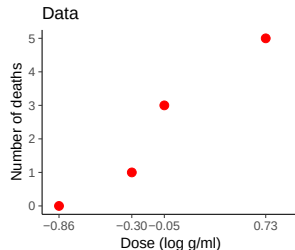
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0.73	5	5



Find out lethal dose 50% (LD50)

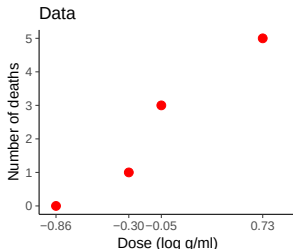
- used to classify how hazardous chemical is
- 1984 EEC directive has 4 levels



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Bayesian methods help to

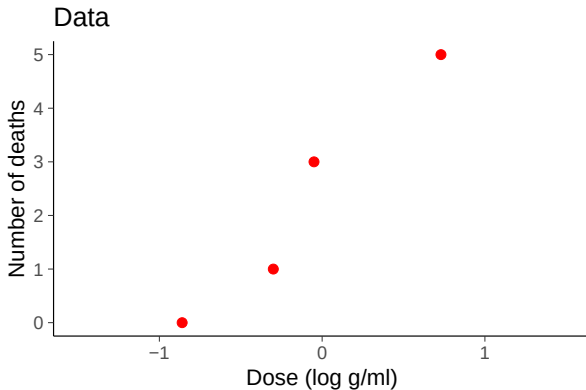
- reduce the number of animals needed
- easy to make sequential experiment and stop as soon as desired accuracy is obtained



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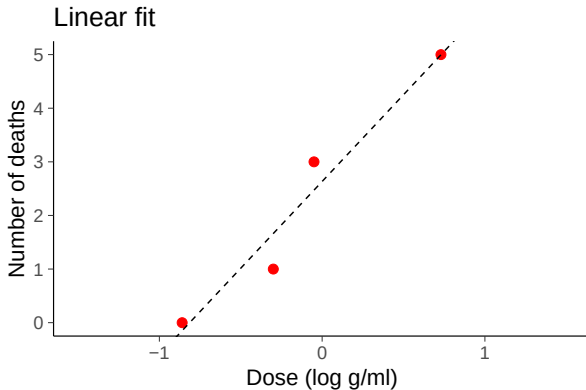
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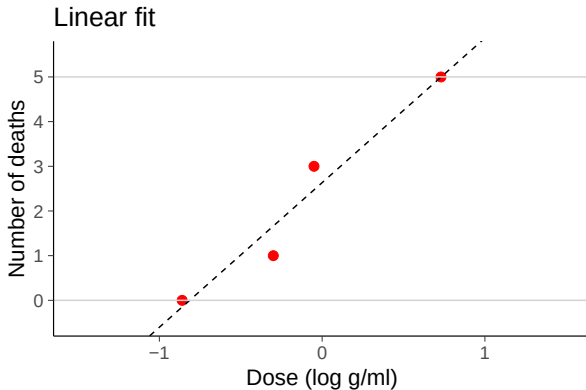


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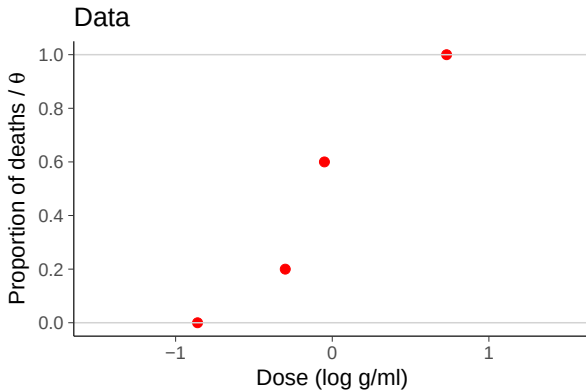
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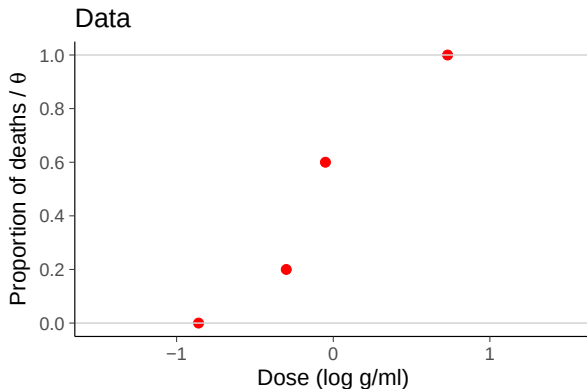
Bioassay





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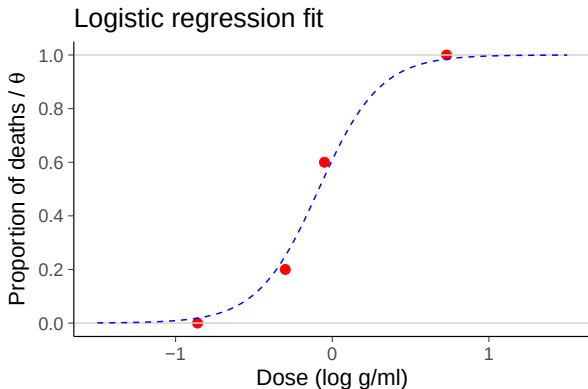
Binomial model

$$y_i \mid \theta_i \sim \text{Bin}(n_i, \theta_i)$$



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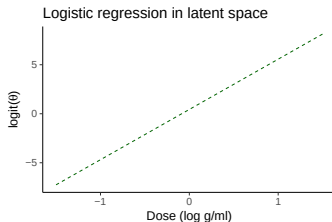
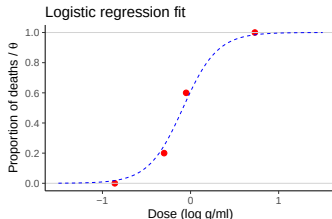
Binomial model

$$y_i \mid \theta_i \sim \text{Bin}(n_i, \theta_i), \quad \text{logit}(\theta_i) = \log\left(\frac{\theta_i}{1 - \theta_i}\right) = \alpha + \beta x_i$$



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$$y_i \mid \theta_i \sim \text{Bin}(n_i, \theta_i)$$
$$\text{logit}(\theta_i) = \log\left(\frac{\theta_i}{1 - \theta_i}\right)$$
$$= \alpha + \beta x_i$$



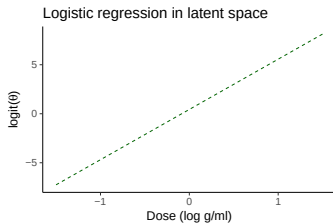
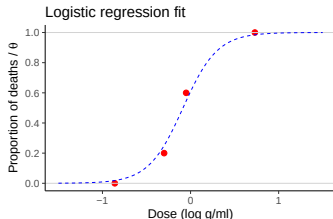


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$$y_i \mid \theta_i \sim \text{Bin}(n_i, \theta_i)$$
$$\text{logit}(\theta_i) = \log\left(\frac{\theta_i}{1 - \theta_i}\right)$$
$$= \alpha + \beta x_i$$

$$\theta_i = \frac{1}{1 + \exp(-(\alpha + \beta x_i))}$$





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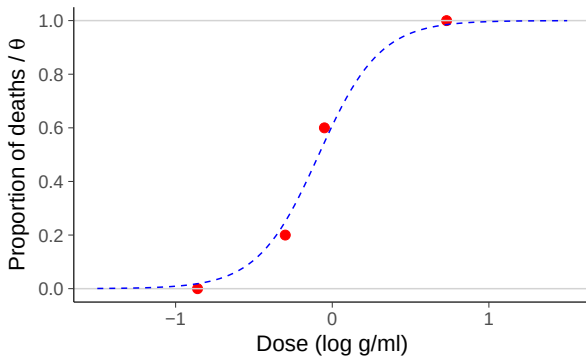
- What assumptions are needed for $y_i \mid \theta_i \sim \text{Bin}(n_i, \theta_i)$?
- Are animals within a dose group exchangeable and independent?
- Name one practical reason why the binomial model could fail.
- Why use a logit model instead of $\theta_i = \alpha + \beta x_i$?



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- In this lecture, we constrain toxicity to be monotone.
 - The grid is restricted to $\beta \geq 0$; increasing dose cannot decrease the probability of death.
 - Conceptually, the LD50 summary is interpreted under the positive-dose-effect assumption $\beta > 0$.
 - If $\beta \leq 0$ were allowed, we would also report $\Pr(\beta > 0 \mid y)$ and LD50 conditional on $\beta > 0$.



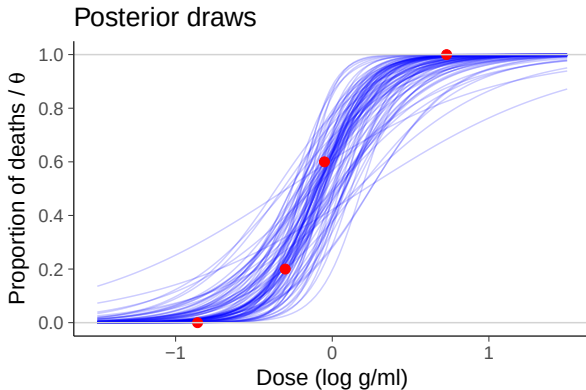
Logistic regression fit



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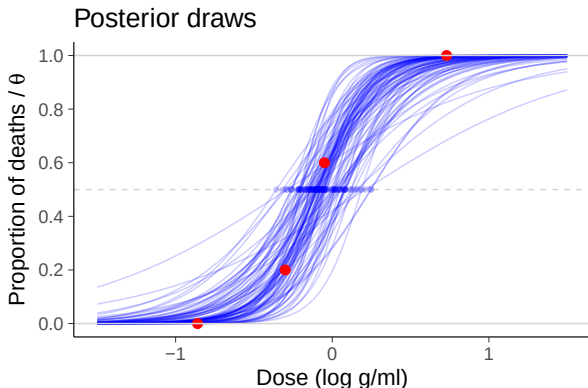
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Bioassay: Lethal Dose 50%

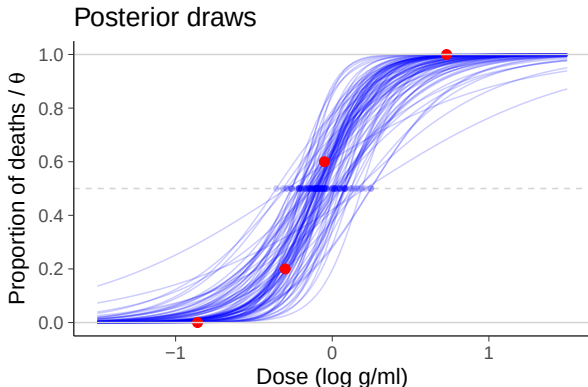


$$E\left(\frac{y}{n}\right) = \text{logit}^{-1}(\alpha + \beta x) = 0.5$$



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Bioassay: Lethal Dose 50%

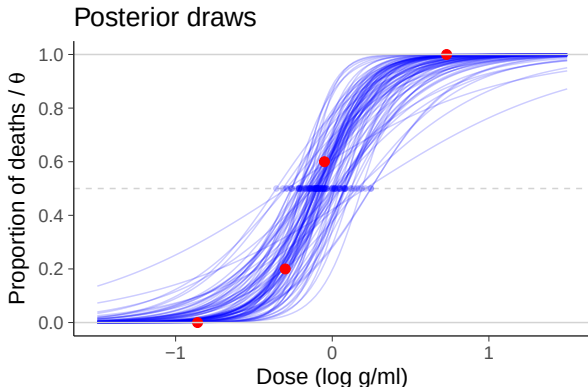


$$E\left(\frac{y}{n}\right) = \text{logit}^{-1}(\alpha + \beta x) = 0.5 \quad \Rightarrow \quad x_{\text{LD50}} = -\alpha/\beta$$



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Bioassay: Lethal Dose 50%

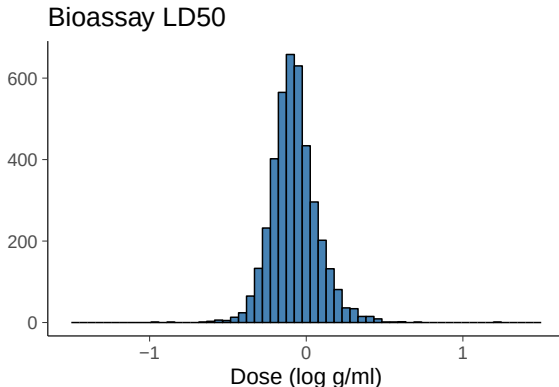


$$E\left(\frac{y}{n}\right) = \text{logit}^{-1}(\alpha + \beta x) = 0.5 \quad \Rightarrow \quad x_{\text{LD50}} = -\alpha/\beta$$
$$x_{\text{LD50}}^{(s)} = -\alpha^{(s)}/\beta^{(s)}$$



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Bioassay: Lethal Dose 50%



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$$x_{\text{LD50}}^{(s)} = -\alpha^{(s)}/\beta^{(s)}$$



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Bioassay posterior

Binomial model

$$y_i \mid \theta_i \sim \text{Bin}(n_i, \theta_i)$$

Link function

$$\text{logit}(\theta_i) = \alpha + \beta x_i$$

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Bioassay posterior

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$$y_i \mid \theta_i \sim \text{Bin}(n_i, \theta_i)$$

Link function

$$\text{logit}(\theta_i) = \alpha + \beta x_i$$

Likelihood

$$p(y_i \mid \alpha, \beta, n_i, x_i) \propto \theta_i^{y_i} [1 - \theta_i]^{n_i - y_i}$$



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Link function

$$\text{logit}(\theta_i) = \alpha + \beta x_i$$

Likelihood

$$\begin{aligned} p(y_i \mid \alpha, \beta, n_i, x_i) &\propto \theta_i^{y_i} [1 - \theta_i]^{n_i - y_i} \\ &\propto [\text{logit}^{-1}(\alpha + \beta x_i)]^{y_i} [1 - \text{logit}^{-1}(\alpha + \beta x_i)]^{n_i - y_i} \end{aligned}$$



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Likelihood

$$\begin{aligned} p(y_i \mid \alpha, \beta, n_i, x_i) &\propto \theta_i^{y_i} [1 - \theta_i]^{n_i - y_i} \\ &\propto [\text{logit}^{-1}(\alpha + \beta x_i)]^{y_i} [1 - \text{logit}^{-1}(\alpha + \beta x_i)]^{n_i - y_i} \end{aligned}$$

Posterior

$$p(\alpha, \beta \mid y, n, x) \propto p(\alpha, \beta) \prod_{i=1}^k p(y_i \mid \alpha, \beta, n_i, x_i)$$

No analytic posterior distribution? What can we do?



For grid point g with parameters (α_g, β_g) :

$$\log \tilde{p}_g = \log p(\alpha_g, \beta_g) + \sum_{i=1}^k \log p(y_i \mid \alpha_g, \beta_g, n_i, x_i)$$

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For grid point g with parameters (α_g, β_g) :

$$\log \tilde{p}_g = \log p(\alpha_g, \beta_g) + \sum_{i=1}^k \log p(y_i \mid \alpha_g, \beta_g, n_i, x_i)$$

$$\hat{p}_g = \frac{\exp(\log \tilde{p}_g)}{\sum_{h=1}^G \exp(\log \tilde{p}_h)}$$



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For grid point g with parameters (α_g, β_g) :

$$\log \tilde{p}_g = \log p(\alpha_g, \beta_g) + \sum_{i=1}^k \log p(y_i \mid \alpha_g, \beta_g, n_i, x_i)$$

$$\hat{p}_g = \frac{\exp(\log \tilde{p}_g)}{\sum_{h=1}^G \exp(\log \tilde{p}_h)}$$

Computing on the log scale avoids numerical underflow and matches the log weights we use later.



1. Set up an area (can be hard) for α and β that captures most mass (here $\alpha \in [-1, 5]$ and $\beta \in [0, 30]$)

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Grid evaluation

1. Set up an area (can be hard) for α and β that captures most mass (here $\alpha \in [-1, 5]$ and $\beta \in [0, 30]$)
2. Compute unnormalized $p(\alpha^{(g)}, \beta^{(g)} \mid y, n, x)$, here $\tilde{p}$, at the grid points g



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Grid evaluation

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3. Sum up $\tilde{p}$ over the whole grid (for all $g \in \{1, \dots, G\}$)



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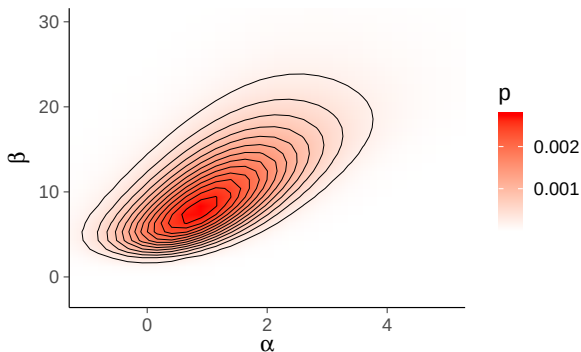
Grid evaluation

1. Set up an area (can be hard) for α and β that captures most mass (here $\alpha \in [-1, 5]$ and $\beta \in [0, 30]$)
2. Compute unnormalized $p(\alpha^{(g)}, \beta^{(g)} \mid y, n, x)$, here $\tilde{p}$, at the grid points g
3. Sum up $\tilde{p}$ over the whole grid (for all $g \in \{1, \dots, G\}$)
4. Compute (normalize) the pmf approximation of the posterior $\hat{p}$

g	(α, β)	$\tilde{p}$	$\hat{p}$
1	$(-1, 0)$	0.02	0.0002
2	$(-0.8, 0.2)$	0.03	0.0003
...	...	...	...
G	$(5, 30)$	0.001	0.00001
$\sum_{g=1}^G$	-	100	1



Posterior density evaluated in a grid

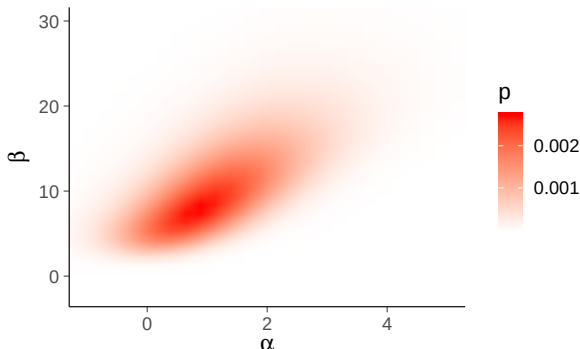


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Bioassay: uniform prior on constrained grid

Posterior density evaluated in a grid



Density evaluated in grid, but plotted using interpolation

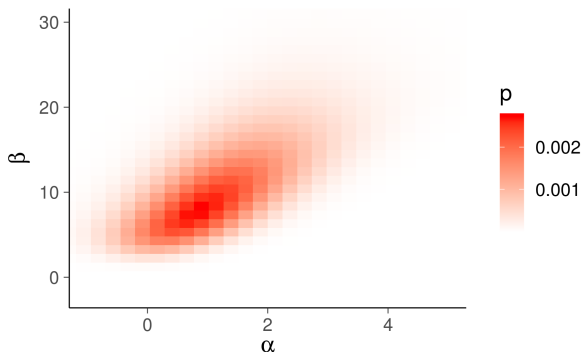
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Bioassay: uniform prior on constrained grid

Posterior density evaluated in a grid

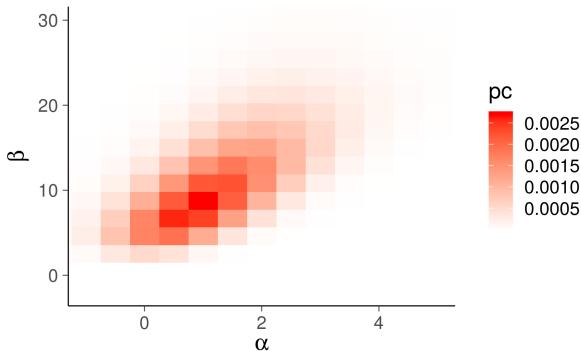


Density evaluated in grid, and plotted without interpolation



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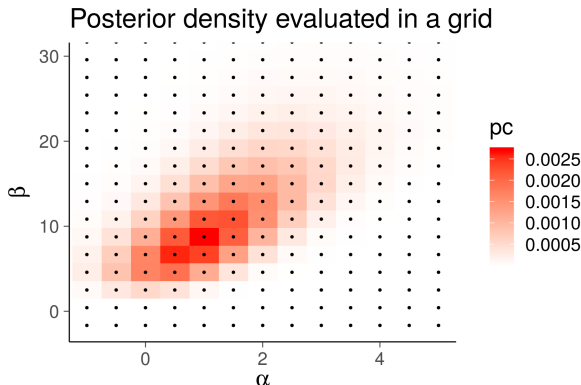
Posterior density evaluated in a grid



Density evaluated in a coarser grid



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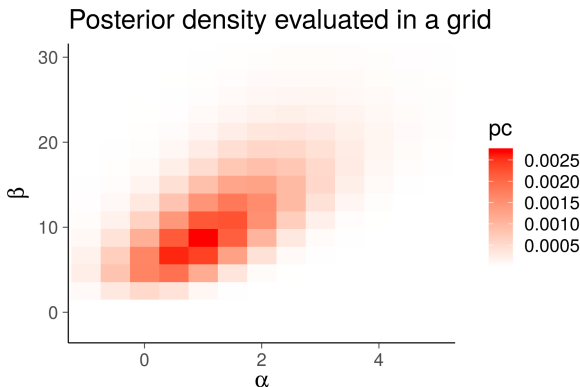


- Approximate the density as piecewise constant function
- Evaluate density in a grid over some finite region
- Density times cell area gives probability mass in each cell



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Bioassay: uniform prior on constrained grid



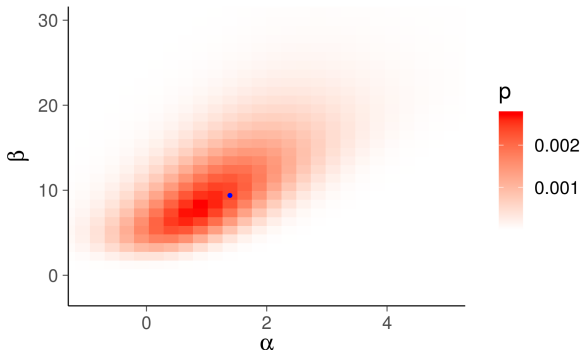
- Densities at 1, 2, and 3: 0.0027 0.0010 0.0001
- Probabilities of cells 1, 2, and 3: 0.0431 0.0166 0.0010
- Probabilities of cells sum to 1



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Bioassay: uniform prior on constrained grid

Posterior density and draws in a grid



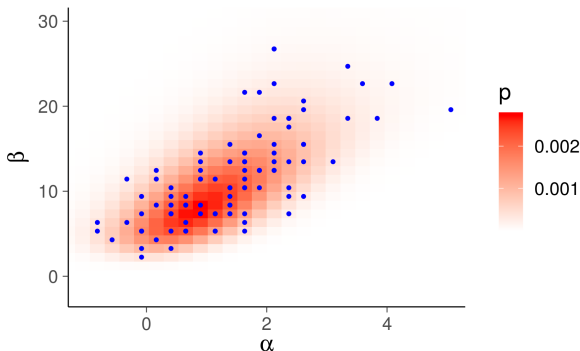
- Sample according to grid cell probabilities



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Posterior density and draws in a grid



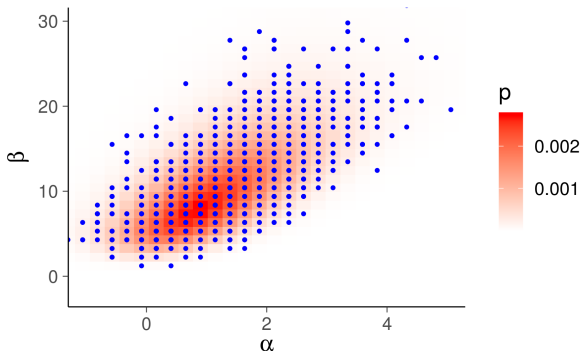
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Bioassay: uniform prior on constrained grid

Posterior density and draws in a grid

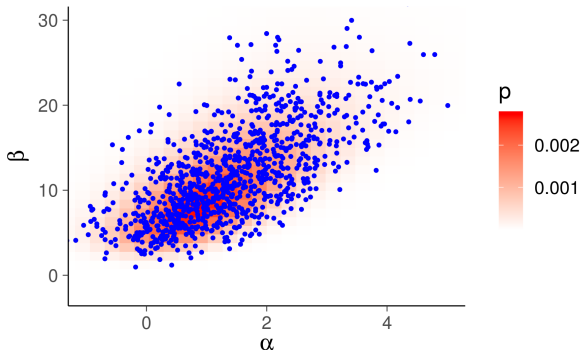


- Sample according to grid cell probabilities
- Several draws can be from the same grid cell



Bioassay: uniform prior on constrained grid

Posterior density in a grid and jittered draws



- Jitter can be added to improve visualization

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- Draws can be used to estimate expectations, for example

$$E[x_{LD50}] = E[-\alpha/\beta] \approx \frac{1}{S} \sum_{s=1}^S \frac{-\alpha^{(s)}}{\beta^{(s)}}$$

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- Draws can be used to estimate expectations, for example

$$E[x_{LD50}] = E[-\alpha/\beta] \approx \frac{1}{S} \sum_{s=1}^S \frac{-\alpha^{(s)}}{\beta^{(s)}}$$

- Instead of sampling, grid could be used to evaluate functions directly, for example

$$E[-\alpha/\beta] \approx \sum_{t=1}^T w_{\text{cell}}^{(t)} \frac{-\alpha^{(t)}}{\beta^{(t)}},$$

where $w_{\text{cell}}^{(t)}$ is the normalized probability of a grid cell t ,
and $\alpha^{(t)}$ and $\beta^{(t)}$ are center locations of grid cells

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- Grid sampling gets computationally too expensive in high dimensions

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- Multiparameter posteriors are joint distributions.
- Marginalization removes nuisance parameters.
- Predictive distributions include parameter uncertainty and new observation noise.
- Posterior draws make derived quantities easy to summarize.
- Bioassay: grid evaluation works here because the parameter dimension is small.
- This motivates more scalable computation for larger Bayesian models.



Multinomial model for categorical data

- Extension of binomial to K categories
- Observation model: **Multinomial model**

$$\mathbf{y} \mid \boldsymbol{\theta}, m \sim \text{Multinomial}(m, \boldsymbol{\theta})$$

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Multinomial model for categorical data

- Extension of binomial to K categories
- Observation model: **Multinomial model**

$$\mathbf{y} \mid \boldsymbol{\theta}, m \sim \text{Multinomial}(m, \boldsymbol{\theta})$$

$$p(\mathbf{y} \mid \boldsymbol{\theta}, m) \propto \prod_{k=1}^K \theta_k^{y_k},$$

$$\text{where } \sum_{k=1}^K \theta_k = 1, \text{ and } \theta_k > 0$$

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Multinomial model for categorical data

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$$\text{where } \sum_{k=1}^K \theta_k = 1, \text{ and } \theta_k > 0$$

- Example: $\mathbf{y} = (0, 1, 0, 6, 30)$ - **what is K here?**



Multinomial model for categorical data

- Extension of binomial to K categories
- Observation model: **Multinomial model**

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$$\text{where } \sum_{k=1}^K \theta_k = 1, \text{ and } \theta_k > 0$$

- Example: $\mathbf{y} = (0, 1, 0, 6, 30)$ - **what is K here?**
- Categorical distribution: $\text{Multinomial}(1, \boldsymbol{\theta})$



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Prior for the multinomial model

- What is important when choosing the prior for θ ?

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- What is important when choosing the prior for θ ?
- Conjugate prior: The Dirichlet distribution

$$p(\theta) \propto \prod_{k=1}^K \theta_k^{\alpha_k - 1},$$

$$\text{where } \alpha_k > 0, \sum_{k=1}^K \theta_k = 1, \text{ and } \theta_k > 0$$

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Prior for the multinomial model

- What is important when choosing the prior for θ ?
- Conjugate prior: The Dirichlet distribution

$$p(\theta) \propto \prod_{k=1}^K \theta_k^{\alpha_k - 1},$$

$$\text{where } \alpha_k > 0, \sum_{k=1}^K \theta_k = 1, \text{ and } \theta_k > 0$$

- What is the connection to the Beta distribution?



Posterior for the multinomial model

Let $y_k = \sum_{i=1}^n y_{k,i}$ be the observed count in category k .

$$p(\boldsymbol{\theta} \mid \mathbf{y}) \propto p(\mathbf{y} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})$$

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$$\propto \prod_{k=1}^K \theta_k^{\alpha_k - 1} \prod_{i=1}^n \prod_{k=1}^K \theta_k^{y_{k,i}}$$

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$$\begin{aligned} &\propto \prod_{k=1}^K \theta_k^{\alpha_k - 1} \prod_{i=1}^n \prod_{k=1}^K \theta_k^{y_{k,i}} \\ &= \prod_{k=1}^K \theta_k^{\alpha_k - 1} \prod_{k=1}^K \theta_k^{\sum_{i=1}^n y_{k,i}} \end{aligned}$$

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- Observation model

$$p(\mathbf{y} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) \propto |\boldsymbol{\Sigma}|^{-1/2} \exp \left(-\frac{1}{2} (\mathbf{y} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \boldsymbol{\mu}) \right),$$

where $\mathbf{y} \in \mathbb{R}^D$

- See BDA3 p. 72.
- New recommended LKJ-prior mentioned in Appendix A, see more in Stan manual